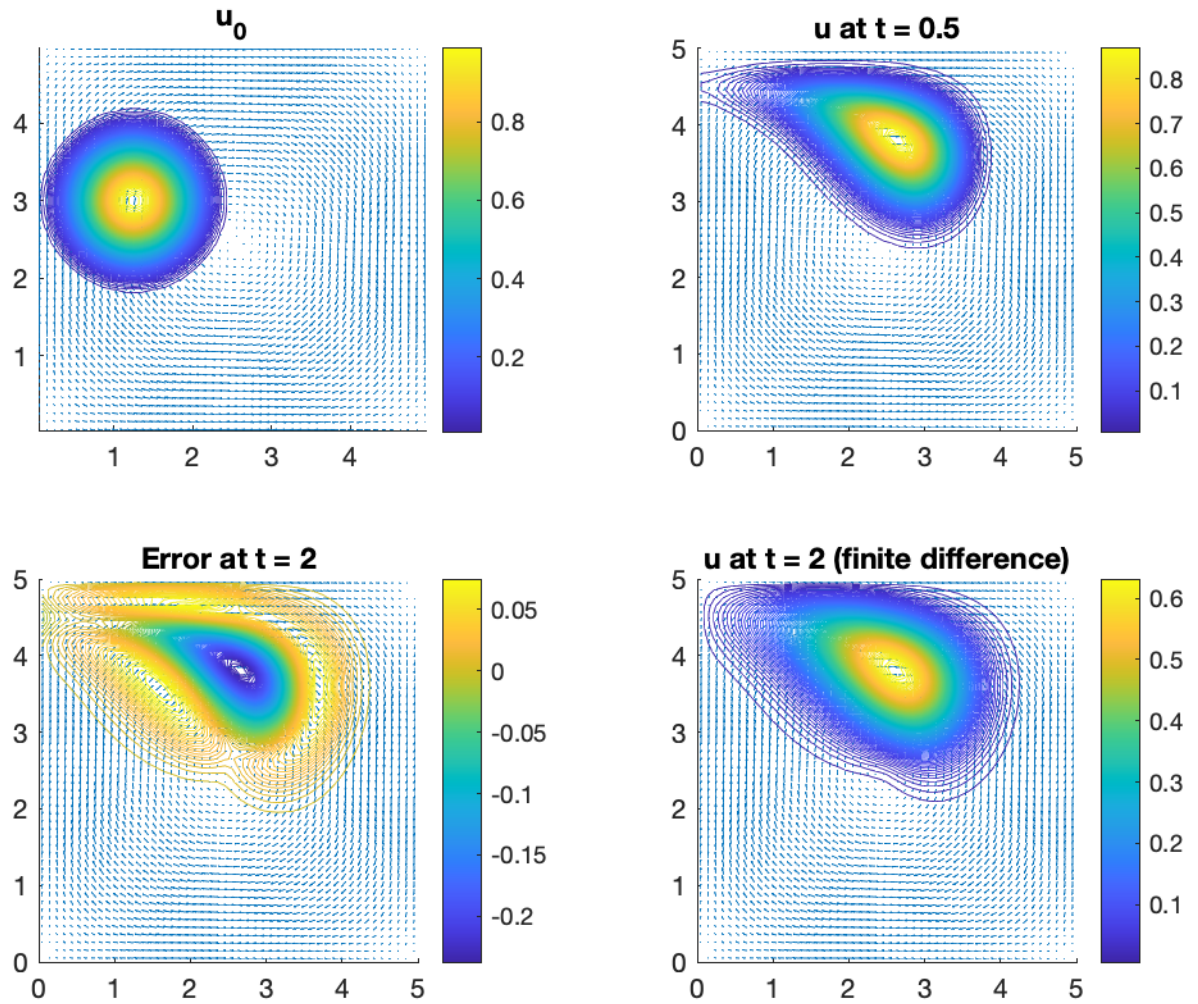


Graphical Abstract

Simulation and maximum likelihood estimation for a stochastic convection-diffusion equation

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Highlights

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- Parameter estimation for the stochastic convection-diffusion equation via MLE.
- Consistent estimators for independent diffusion and advective coupling parameters.
- Asymptotic normality is proved using Malliavin-stein method.
- Spectral analysis resolves the interplay between deterministic transport and noise.
- Numerical simulations via spectral Galerkin methods validate the MLE framework.

Simulation and maximum likelihood estimation for a stochastic convection-diffusion equation

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Abstract

This paper addresses the parameter estimation problem for a Stochastic Partial Differential Equation (SPDE); more precisely, for a stochastic convection-diffusion equation driven by infinite-dimensional Gaussian noise. Moving beyond traditional bidiagonal frameworks where spectral modes evolve independently, the proposed model incorporates an incompressible velocity field that induces modal coupling within the Galerkin projection. We derive Maximum Likelihood Estimators (MLE) for both the diffusion and convection coefficients based on continuous observations of the first N Fourier modes. Under suitable regularity conditions, we rigorously prove the strong consistency of the estimators and establish their asymptotic normality via Malliavin-Stein method. Finally, the performance of the method is validated through numerical simulations, confirming the stability and accuracy of the estimators in physical systems characterized by non-diagonal convective interactions.

Keywords: Stochastic partial differential equations, Convection-diffusion equation, Parameter estimation, Spectral Galerkin method, Malliavin-Stein method

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1. Introduction and main contribution

The deterministic convection-diffusion equation is a fundamental partial differential equation describing the transport of particles, energy, or other physical quantities within a system via two intertwined mechanisms: diffusion and advection. This deterministic framework has been extensively applied across various scientific and engineering disciplines. Prominent applications include the modeling of pollutant dispersion in environmental engineering, heat transfer in fluid dynamics, and complex biological and chemical processes, such as the dynamics of bubble growth in organic media [1, 2].

However, in many realistic macroscopic scenarios, these dynamical systems are subjected to random perturbations and uncertainties arising from thermal fluctuations, turbulent fluxes, microscopic heterogeneities, or unpredictable environmental noise. To rigorously account for these effects, the classical deterministic model is naturally extended to a Stochastic Partial Differential Equation (SPDE). Among these, the stochastic convection-diffusion equation is of particular interest, as it governs transport within a fluid medium influenced by both deterministic flows and stochastic forcing.

In this paper, we study the statistical properties and numerical simulation of the following stochastic convection-diffusion equation in a bounded spatial domain $\mathcal{O} \subset \mathbb{R}^d$:

$$du = [\beta \Delta u - \theta \bar{v} \cdot \nabla u] dt + \sigma \sum_{k \in \mathcal{J}} \lambda_k^{-\gamma} h_k dB_k(t). \quad (1)$$

Here, the process $u(t, x)$ models the stochastically fluctuating concentration of a passive tracer advected by an incompressible fluid. Consequently, the velocity field $\bar{v}(x)$ is divergence-free ($\nabla \cdot \bar{v} = 0$) and, given the passive nature of the tracer, it exerts no dynamic feedback on the fluid. This decoupling allows us to treat the velocity field $\bar{v}(x)$ as an independently given coefficient rather than part of a coupled Navier-Stokes system.

Our framework can easily be applied to time-dependent velocity fields $\bar{v}(t, x)$. However, to keep the presentation simple, we assume the velocity field only varies in space. Far from being a mere mathematical convenience, this stationary assumption is physically well-justified for numerous macroscopic transport phenomena. For instance, extensive oceanographic modeling by Zavala-Hidalgo et al. [3, 4] demonstrates that large-scale marine dynamics in the Gulf of Mexico—such as wind-driven coastal currents and the Loop Current—often maintain a remarkably constant directional flow over prolonged seasonal periods spanning several months. Under such quasi-steady regimes, a time-independent advection field accurately captures the sustained deterministic transport of passive masses. This makes our model highly suitable for real-world ecological and environmental applications, such as tracking the long-term dispersion and coastal arrival of *Sargassum* macroalgae to the Yucatan Peninsula. Thus, focusing on a stationary velocity field not only reflects realistic, long-duration environmental conditions, but also helps us clearly understand the analytical interaction between the deterministic transport operators and the stochastic noise.

We supplement equation (1) with appropriate initial data and homogeneous Neumann boundary conditions (zero-flux) for the tracer. To prevent fluid transport across the boundary, we impose a standard no-penetration (slip) condition on the velocity field, ensuring that its normal component vanishes on the boundary. It is worth noting that the analytical and numerical frameworks developed herein can be naturally extended to other boundary configurations.

Mathematically, (1) constitutes a linear SPDE with spatially varying coefficients. The driving stochastic forcing is an infinite-dimensional, Gaussian noise, white in time and colored in space, rigorously formulated via a sequence of independent standard Wiener processes $\{B_k(t)\}_{k \geq 1}$. The fractional powers of the eigenvalues, $\lambda_k^{-\gamma}$ for $\gamma > 0$, act as a spectral dampening mechanism, ensuring the noise is spatially colored and the corresponding covariance operator has finite trace.

A critical challenge in the empirical application of transport models is the accurate specification of the underlying velocity field $\bar{v}$. As we already mention, in our framework we treat $\bar{v}$ as a spatial vector field that prescribes the geometry of the flow, while we introduce the parameter $\theta > 0$ to act as an advective coupling constant. From a practical modeling perspective, θ acts as a crucial corrector that absorbs unresolved temporal fluctuations, small-scale turbulent eddies, or inherent uncertainties in the macroscopic velocity measurements.

Physically, θ can be interpreted as an effective medium coefficient [5]. While classical fluid dynamics often reduces the advection-diffusion equation to a single dimensionless parameter (such as the Péclet number) [1], maintaining two independent parameters— β for the Laplacian and θ for the convective term—grants the model significantly more flexibility to adapt to complex empirical data. For instance, in porous media or biological aggregation models [6], the tracer velocity rarely matches the macroscopic fluid velocity identically (the $\theta = 1$ case) due to physical retardation factors or active biological advection coefficients. Thus, the model given by (1) serves as a robust generalization of the fully aligned case.

The mathematical treatment of passive scalar advection has long been a focal point in statistical physics and turbulence theory, most notably within the Kraichnan model framework. In this setting, the interaction between a stochastic velocity field and a scalar concentration is known to produce complex phenomenological features, such as anomalous scaling and intermittency. For instance, Andreanov, Komarova, and Nalimov [7] explored these dynamics by employing large-order asymptotics of quantum-field-theoretic expansions to characterize the behavior of the advective coupling. While their research focuses on the perturbative analysis of the scalar's fluctuations, it underscores the fundamental importance of the advection operator $\bar{v} \cdot \nabla u$ and its associated coupling parameters in governing transport phenomena. Our work bridges the gap between these theoretical physics foundations and statistical methodology by addressing the inverse problem: rather than analyzing scaling exponents, we utilize the spectral properties of this advective interaction to derive consistent parameter estimators for the independent physical intensities β and θ .

1.1. Our contribution

The main contributions of this work are twofold. First, we perform high-fidelity simulations of the SPDE driven by an empirical divergence-free vector field. By leveraging the spectral decomposition of the solution and projecting the SPDE using an orthonormal basis $\{h_k\}$ of $\mathcal{H}$ (to be defined latter on), we transform the system into an infinite-dimensional, coupled system of Ornstein-Uhlenbeck-type SDEs. **Comentario: todavía no definimos la base ortonormal ni el espacio $\{h_k\}$.**

Second, we address the inverse problem: the statistical estimation of the physical parameters. We propose Maximum Likelihood Estimators (MLE) for the diffusion coefficient β and the convective coupling parameter θ based on a finite horizon of *continuous* observations of the Fourier modes. Using the Girsanov theorem for finite-dimensional systems, we derive closed-form expressions for the estimators [8]. Subsequently, we rigorously prove that these estimators are strongly consistent as both the number of spectral modes N and the time horizon T simultaneously tend to infinity. Furthermore, by employing the Malliavin-Stein approach [9], we establish their asymptotic normality under this same joint limit, providing a robust theoretical foundation for statistical inference in stochastically forced fluid models.

1.2. Statistical Motivation: Dynamic Noise vs. Measurement Error

In the classical framework of deterministic partial differential equations, parameter estimation is typically addressed through the lens of Ordinary or Nonlinear Least Squares (LS). In such settings, the underlying physical process is assumed to follow a rigid deterministic law, and any discrepancy between the model and the observations is attributed to *measurement error* or *extrinsic noise*. Specifically, if $u(t, x; \theta)$ denotes the solution to a deterministic convection-diffusion equation, the observed data y_{obs} is modeled as $y_{obs} = u(t, x; \theta) + \epsilon$, where ϵ represents a Gaussian white noise independent of the physical dynamics.

In contrast, the SPDE framework treats noise as an *intrinsic* or *dynamic* component of the system. Here, the stochastic forcing σdW is not a nuisance added during the observation process, but rather a driving force that interacts with the diffusion (β) and convection (θ) operators at every infinitesimal time step. This distinction is mathematically profound: while LS methods focus on the mean behavior and treat fluctuations as residuals to be minimized, the Maximum Likelihood Estimator (MLE) for SPDEs leverages the structure of these fluctuations as a primary source of information [10, 8].

By analyzing the trajectory of the spectral modes, the MLE accounts for the temporal and spatial correlations induced by a coupling matrix (see Section 2 for the full details). **Todavía no se define A.** In a deterministic LS approach, the "noise" is typically assumed to be independent and identically distributed (i.i.d.), effectively ignoring the physical constraints that the fluid flow imposes on the dispersion of the scalar. However, in our SPDE framework, the Girsanov-based likelihood function captures how the convective field $\bar{v}$ "shapes" the noise. This allows

for more robust inference, particularly in high-dimensional systems where the variability of the data—often discarded as experimental error in deterministic models—contains essential signatures of the underlying transport coefficients [11, 12].

In recent years, significant progress has been made in the statistical inference for semilinear SPDEs. For instance, Pasemann and Stannat [13] established a comprehensive framework for drift estimation in stochastic reaction-diffusion systems, providing rigorous conditions for consistency and asymptotic normality of Maximum Likelihood Estimators (MLE) as the number of spectral modes increases. Their work particularly emphasizes the robustness of these estimators under model misspecification within the reaction terms. While their approach primarily addresses systems where the dominant operator is often diagonal in the spectral domain, our work extends the inferential challenge to convection-diffusion SPDEs. In our setting, the velocity field $\bar{v}$ introduces a non-trivial coupling between Fourier modes through the advection operator, leading to a skew-symmetric interaction matrix that necessitates a distinct treatment of the likelihood function and its asymptotic properties.

In this work, we test the limits of the spectral estimation framework originally pioneered by Huebner, Lototsky and Rozovsky in the 1990s [14] and further refined by Lototsky [15], (see Cialenco [8] for a good review of the topic), and [16]. Traditionally, the existing literature has focused on what are often termed "bidiagonal" SPDEs, where the underlying operators share an orthonormal basis, resulting in a system of independent stochastic differential equations (SDEs) for the spectral modes u_k . While such decoupling allows for straightforward statistical analysis and efficient simulation, it fails to capture the intricate dynamics of many physical transport phenomena. In contrast, our model involves an infinite system of coupled SDEs, driven by a convective velocity field that necessitates simultaneous truncation and resolution of the entire system. Despite the significant analytical and computational challenges posed by this modal coupling, we demonstrate that it is still possible to rigorously establish the statistical properties of the estimators and successfully perform parameter inference. To the best of our knowledge, this is the first time the spectral method has been applied to an SPDE characterized by such full modal interaction, extending its reach beyond the traditional decoupled framework.

The paper is organized as follows.

2. Framework

Set a bounded and smooth enough domain $O \subset \mathbb{R}^d$ and define $\mathcal{H} = L^2(O)$.

It is known (see [17] or Section XIII.15 in [18]) that the set of eigenvalues and eigenvectors of Δ

$$-\Delta h_k = \lambda_k h_k$$

satisfies:

- a) $\{h_k\}_{k \in \mathbb{N}}$ forms a complete orthonormal system in $L^2(O)$.

- b) $\{\lambda_k\}_{k \in \mathbb{N}}$, can be arranged such that $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots$ with $\lambda_k \rightarrow +\infty$ as $k \rightarrow +\infty$.

In addition, it is also known that for domains with sufficiently regular boundaries there exists a positive constant ϖ so that

$$\lim_{k \rightarrow \infty} |\lambda_k| k^{-2/d} = \varpi. \quad (2)$$

Consider a stochastic basis $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ that satisfies the usual assumptions, that is, $(\Omega, \mathcal{F}, \mathbb{P})$ is a complete probability space and $\{\mathcal{F}_t\}_{t \geq 0}$ is a filtration satisfying the usual hypothesis. On this basis, we denote by $\{w_j, j \geq 1\}$ a collection of independent standard Wiener processes.

The full model we will consider is the equation (1) with the following assumptions

$$\begin{cases} du = \beta \Delta u dt - \theta \bar{v} \cdot \nabla u dt \\ \quad + \sigma \sum_{\mathbf{k} \in \mathcal{J}} \lambda_{\mathbf{k}}^{-\gamma} h_{\mathbf{k}}(x) dw_{\mathbf{k}}(t), & \text{in } (0, +\infty) \times O, \\ \frac{\partial u}{\partial \eta} = 0, & \text{on } (0, +\infty) \times \partial O, \\ u(0, x) = u_0(x), & \text{in } O, \end{cases} \quad (3)$$

where $\theta > 0, \beta > 0$ are parameters to be estimated. In addition, $\gamma \geq 0, \sigma > 0$, and $u_0 \in L^2(O)$, with $\mathbf{k}$ is an index belonging to $\mathcal{J}$. Here $\mathcal{J}$ is a index set compose of the basis for the ilbert space $\mathcal{H}$.

We interpret u in (3) as the *stochastic* concentration for mass transfer, or the temperature for heat transfer (depending of the actual application), under the influence of random external forces. The concentration is subject to two types of dispersals; meaning the term with the gradient represents the convection of a scalar along a flow field, while the term with the Laplacian describes the classical diffusion.

It is not difficult to show that equation (3) has a unique adapted variational strong solution u in probability sense (see Section 4.4.1 in [12]), such that

$$u \in L^2(\Omega; C((0, T); L^2(O))). \quad (4)$$

We set $u_{\mathbf{k}}$ as

$$u_{\mathbf{k}}(t) = (u(t, \cdot), h_{\mathbf{k}}(\cdot))_{\mathcal{H}}, \quad \mathbf{k} \in \mathcal{J},$$

this means, $u_{\mathbf{k}}$ is the Fourier coefficient of the solution u of (3) with respect to $h_{\mathbf{k}}, \mathbf{k} \in \mathcal{J}$. From (4), we could write the solution u as

$$u(t, x) = \sum_{\mathbf{k} \in \mathcal{J}} u_{\mathbf{k}}(t) h_{\mathbf{k}}(x) \quad (5)$$

It is not difficult to see that the every $u_{\mathbf{k}}$ follow the dynamics of a Ornstein-Uhlenbeck (OU) process in $\mathbb{R}^N$. Indeed, following a classical procedure we could arrive to the following SDE for $u_{\mathbf{k}}$,

$$du_{\mathbf{k}} = -\beta \lambda_{\mathbf{k}} u_{\mathbf{k}}(t) dt - \theta \sum_{\mathbf{j} \in \mathcal{J}} u_{\mathbf{j}}(t) v_{\mathbf{j}, \mathbf{k}} dt + \sigma \lambda_{\mathbf{k}}^{-\gamma} dw_{\mathbf{k}}(t), \quad (6)$$

$$u_{\mathbf{k}}(0) = (u_0, h_{\mathbf{k}}), \quad t \geq 0,$$

where,

$$v_{j,k} := (\bar{v} \cdot \nabla h_j, h_k(\cdot))_{\mathcal{H}}. \quad (7)$$

Observe that in general $v_{j,k} \neq v_{k,j}$, and in our particular case we have that $v_{j,k} = -v_{k,j}$ (see appendix for the proof). Thus, u_k solves an infinite dimensional coupled system of SDEs given by the OU processes.

We define a truncation of the index set $\mathcal{J}$ as

$\mathcal{J}^N :=$ a finite subset of N members of the elements of the basis.

We could assume that $\lim_{N \rightarrow \infty} \mathcal{J}^N = \mathcal{J}$.

We truncate the system (6) up to level $\mathcal{J}^N$:

$$du_k = -\beta \lambda_k u_k(t) dt - \theta \sum_{j \in \mathcal{J}^N} u_j(t) v_{j,k} dt + \sigma \lambda_k^{-\gamma} dw_k(t), \quad (8)$$

$$u_k(0) = (u_0, h_k), \quad t \geq 0.$$

Set H^N the finite-dimensional subspace of $L^2(\mathcal{O})$ generated by the first N elements of the basis: $\{h_k\}_{k \in \mathcal{J}^N}$. Moreover, denote by P_N the projection operator of $L^2(\mathcal{O})$ into H^N and define $U^N = P_N u$.

We define the stochastic vector $U = u^N = (u_1, \dots, u_N)$.

Thus, we could rewrite the system (8) as

$$\begin{aligned} dU &= -\beta \Lambda U dt - \theta A U dt + \sigma B dW(t) \\ &= (-\beta \Lambda - \theta A) U dt + \sigma B dW(t) \end{aligned} \quad (9)$$

where A and B are matrix, such that B is diagonal, and A is skew-symmetric (see appendix for further properties of the matrix A). In fact, the (j, k) entry of A is equal to $v_{j,k}$ and for B we have that $(B)_{k,k} = \lambda_k^{-\gamma}$ and $(B)_{j,k} = 0$ when $j \neq k$.

Furthermore, Λ is a diagonal matrix with the $(\Lambda)_{k,k}$ entry equals to λ_k , and $W = (w_1, \dots, w_N)$.

It is known that there is a strong solution for U (see [19] for instance). Indeed, its solution could be represented as

$$U(t) = e^{-\mathcal{A}t} \left[U^0 + \int_0^t e^{\mathcal{A}r} B dW(r) \right] \quad (10)$$

where U^0 is the initial values of the vector U , $\mathcal{A} = \beta \Lambda + \theta A$. In addition, for any $s \geq 0$, $t \geq 0$, it has mean and covariance given by

$$\mathbb{E}[U(t)] = e^{-\mathcal{A}t} U^0 \quad (11)$$

$$\mathbb{E}[U(t)U^T(s)] = e^{-\mathcal{A}t} \left[\int_0^{t \wedge s} e^{\mathcal{A}r} B B^T e^{\mathcal{A}^T r} dr \right] e^{-\mathcal{A}^T s} \quad (12)$$

From this we deduce the variance

$$\text{Var}(U(t)) = e^{-\mathcal{A}t} \left[\int_0^t e^{\mathcal{A}r} B B^T e^{\mathcal{A}^T r} dr \right] e^{-\mathcal{A}^T t}. \quad (13)$$

and its second moment:

$$\mathbb{E}(U^2(t)) = e^{-\mathcal{A}t} \left[\int_0^t e^{\mathcal{A}r} B B^T e^{\mathcal{A}^T r} dr \right] e^{-\mathcal{A}^T t} + e^{-\mathcal{A}t} U^0 (U^0)^T e^{-\mathcal{A}^T t}. \quad (14)$$

Furthermore, for any $s \geq 0$, $t \geq 0$ the distribution of $U(t)$ conditioned on $U(s)$ is Gaussian:

$$U(t) | U(s) \sim \mathcal{N}(e^{-\mathcal{A}(t-s)} U(s), \int_0^{t-s} e^{\mathcal{A}r} B B^T e^{\mathcal{A}^T r} dr)$$

2.1. Convergence Analysis of the Galerkin Approximation

To rigorously establish the convergence of the truncated finite-dimensional system $u^N(t) := U^N(t)$ to the infinite-dimensional mild solution $u(t)$, we analyze the evolution of the approximation error in the $L^2(\Omega; \mathcal{H})$ norm. We define the total error at time t as:

$$e^N(t) := u(t) - u^N(t). \quad (15)$$

A fundamental challenge in analyzing this error stems from the fact that the orthogonal projection operator onto the finite-dimensional subspace $\mathcal{H}_N = \text{span}\{h_1, \dots, h_N\}$, denoted as P_N , does not commute with the advection operator $\bar{v} \cdot \nabla$. This non-commutativity arises because the spatially dependent velocity field $\bar{v}(x)$ continuously transfers energy across different Fourier modes, as characterized by the dense coupling matrix A .

To isolate the truncation effects, we decompose the total error into two orthogonal components:

$$e^N(t) = \rho^N(t) + \delta^N(t), \quad (16)$$

where $\delta^N(t) = (I - P_N)u(t)$ represents the infinite-dimensional tail (the projection error), and $\rho^N(t) = P_N u(t) - u^N(t)$ represents the Galerkin error strictly confined within the finite-dimensional subspace $\mathcal{H}_N$.

Step 1: Evolution of the Galerkin Error $\rho^N(t)$

We begin by deriving the stochastic differential equation governing ρ^N . Applying the projection operator P_N to the original SPDE and subtracting the truncated equation for u^N , we obtain:

$$\begin{aligned} d\rho^N &= [\beta \Delta \rho^N - \theta P_N(\bar{v} \cdot \nabla u) + \theta P_N(\bar{v} \cdot \nabla u^N)] dt \\ &\quad + (P_N B dW(t) - B^N dW^N(t)). \end{aligned} \quad (17)$$

Since the driving noise in the truncated system is exactly the projection of the infinite-dimensional noise (i.e., $B^N dW^N(t) \equiv P_N B dW(t)$), the stochastic differentials cancel out perfectly. Consequently, ρ^N evolves pathwise as a deterministic, differentiable process.

Applying the chain rule (or equivalently, Itô's formula with zero quadratic variation) to the squared L^2 -norm $\|\rho^N(t)\|_{\mathcal{H}}^2$, we get:

$$\frac{1}{2} \frac{d}{dt} \|\rho^N\|^2 = \beta \langle \Delta \rho^N, \rho^N \rangle_{\mathcal{H}} - \theta \langle P_N(\bar{v} \cdot \nabla(u - u^N)), \rho^N \rangle_{\mathcal{H}}. \quad (18)$$

Step 2: Resolving the Operators

We now meticulously resolve the two inner products on the right-hand side of Equation (18).

The Diffusive Term: Assuming homogeneous Dirichlet boundary conditions (or suitable periodic/Neumann conditions), we

apply Green's first identity (integration by parts). The boundary integral vanishes, yielding:

$$\beta \langle \Delta \rho^N, \rho^N \rangle_{\mathcal{H}} = -\beta \langle \nabla \rho^N, \nabla \rho^N \rangle_{\mathcal{H}} = -\beta \|\nabla \rho^N\|_{\mathcal{H}}^2. \quad (19)$$

This confirms the strictly dissipative nature of the Laplacian operator.

The Advective Term: For the second inner product, we use the properties of the orthogonal projector P_N in the Hilbert space $\mathcal{H}$. Because P_N is self-adjoint, we can move it across the inner product:

$$\langle P_N(\bar{v} \cdot \nabla(u - u^N)), \rho^N \rangle_{\mathcal{H}} = \langle \bar{v} \cdot \nabla(u - u^N), P_N \rho^N \rangle_{\mathcal{H}}. \quad (20)$$

By definition, $\rho^N \in \mathcal{H}_N$, which implies that projecting it onto $\mathcal{H}_N$ leaves it invariant ($P_N \rho^N = \rho^N$). Thus, the projector drops out. Substituting the error decomposition $u - u^N = \rho^N + \delta^N$, we split the term using the linearity of the inner product:

$$\langle \bar{v} \cdot \nabla(\rho^N + \delta^N), \rho^N \rangle_{\mathcal{H}} = \langle \bar{v} \cdot \nabla \rho^N, \rho^N \rangle_{\mathcal{H}} + \langle \bar{v} \cdot \nabla \delta^N, \rho^N \rangle_{\mathcal{H}}. \quad (21)$$

We now encounter the critical skew-symmetry property of the convective operator. Since the velocity field is incompressible ($\nabla \cdot \bar{v} = 0$), the divergence theorem ensures that the self-coupling term is identically zero:

$$\langle \bar{v} \cdot \nabla \rho^N, \rho^N \rangle_{\mathcal{H}} = \frac{1}{2} \int_{\mathcal{O}} \bar{v} \cdot \nabla |\rho^N|^2 dx = -\frac{1}{2} \int_{\mathcal{O}} (\nabla \cdot \bar{v}) |\rho^N|^2 dx = 0. \quad (22)$$

Step 3: Bounding the Galerkin Error

Substituting these resolved terms back into Equation (18), the energy evolution simplifies significantly:

$$\frac{1}{2} \frac{d}{dt} \|\rho^N\|^2 = -\beta \|\nabla \rho^N\|^2 - \theta \langle \bar{v} \cdot \nabla \delta^N, \rho^N \rangle_{\mathcal{H}}. \quad (23)$$

To control the remaining cross-term, we first apply integration by parts. Utilizing the incompressibility of the velocity field ($\nabla \cdot \bar{v} = 0$) and the boundary conditions, we transfer the spatial derivative from the infinite-dimensional tail δ^N to the finite-dimensional error ρ^N :

$$\theta \langle \bar{v} \cdot \nabla \delta^N, \rho^N \rangle_{\mathcal{H}} = -\theta \langle \delta^N, \bar{v} \cdot \nabla \rho^N \rangle_{\mathcal{H}}. \quad (24)$$

Taking the absolute value, we apply the Cauchy-Schwarz inequality, bounding the velocity field by its supremum norm:

$$|\theta \langle \delta^N, \bar{v} \cdot \nabla \rho^N \rangle_{\mathcal{H}}| \leq \theta \|\bar{v}\|_{\infty} \|\delta^N\| \|\nabla \rho^N\|. \quad (25)$$

At this stage, we apply Young's inequality with a free parameter $\epsilon > 0$ (i.e., $ab \leq \frac{\epsilon}{2} a^2 + \frac{1}{2\epsilon} b^2$). By strategically choosing $\epsilon = \beta$, we can decouple the product into:

$$\leq \frac{\beta}{2} \|\nabla \rho^N\|^2 + \frac{\theta^2 \|\bar{v}\|_{\infty}^2}{2\beta} \|\delta^N\|^2. \quad (26)$$

Substituting this bound back into the energy evolution equation for ρ^N , we obtain:

$$\frac{1}{2} \frac{d}{dt} \|\rho^N\|^2 \leq -\beta \|\nabla \rho^N\|^2 + \frac{\beta}{2} \|\nabla \rho^N\|^2 + C \|\delta^N\|^2 = -\frac{\beta}{2} \|\nabla \rho^N\|^2 + C \|\delta^N\|^2, \quad (27)$$

where $C = \frac{\theta^2 \|\bar{v}\|_{\infty}^2}{2\beta}$. The strictly negative term $-\frac{\beta}{2} \|\nabla \rho^N\|^2$ provides additional dissipation, ensuring that the growth of the Galerkin error $\rho^N(t)$ is entirely controlled by the projection error $\delta^N(t)$.

Consequently, to determine the overall convergence rate, it suffices to analyze the asymptotic decay of the infinite-dimensional tail $\delta^N(t)$.

Step 4: Asymptotic Decay of the Tail

To explicitly compute the variance of the infinite-dimensional tail $\delta^N(t)$, we rely on the mild solution representation of the SPDE and Parseval's identity. Assuming a zero initial condition ($u_0 = 0$) for simplicity, the continuous mild solution in the Hilbert space $\mathcal{H}$ is given by the stochastic convolution:

$$u(t) = \int_0^t e^{-\mathcal{A}(t-s)} \sigma B dW(s), \quad (28)$$

where $e^{-\mathcal{A}t}$ is the strongly continuous semigroup generated by the operator $\mathcal{A} = \beta \Lambda + \theta A$, and B is the covariance operator acting on the basis elements such that $B h_{\mathbf{k}} = \lambda_{\mathbf{k}}^{-\gamma} h_{\mathbf{k}}$.

The tail of the Galerkin approximation is obtained by applying the orthogonal projector associated with the complement of the finite-dimensional subspace $\mathcal{H}_N$, denoted as $I - P_N$. By Parseval's identity, the squared L^2 -norm of this tail can be exactly expressed as the infinite sum of its squared spectral components:

$$\|\delta^N(t)\|_{\mathcal{H}}^2 = \|(I - P_N)u(t)\|_{\mathcal{H}}^2 = \sum_{\mathbf{k} \notin \mathcal{J}^N} |u_{\mathbf{k}}(t)|^2. \quad (29)$$

Taking the expected value, we apply the infinite-dimensional Itô isometry to the stochastic convolution. Since the sequence of driving Wiener processes $\{w_{\mathbf{k}}\}_{\mathbf{k} \geq 1}$ are mutually independent, the cross-variation terms vanish. Consequently, the expectation of the squared norm simplifies to the sum of the integrated variances of the individual modes:

$$\mathbb{E} \|\delta^N(t)\|_{\mathcal{H}}^2 = \sigma^2 \sum_{\mathbf{k} \notin \mathcal{J}^N} \int_0^t \|e^{-\mathcal{A}(t-s)} B h_{\mathbf{k}}\|_{\mathcal{H}}^2 ds. \quad (30)$$

Finally, substituting the explicit eigenvalue action of the covariance operator B onto the basis functions, we obtain the rigorous formulation for the expected energy of the unresolved high-frequency modes:

$$\mathbb{E} \|\delta^N(t)\|_{\mathcal{H}}^2 = \sigma^2 \sum_{\mathbf{k} \notin \mathcal{J}^N} \int_0^t \|e^{-\mathcal{A}(t-s)} h_{\mathbf{k}}\|_{\mathcal{H}}^2 \lambda_{\mathbf{k}}^{-2\gamma} ds. \quad (31)$$

Because the convective matrix A is skew-symmetric, it does not contribute to exponential growth; the semigroup contraction is governed by the smallest eigenvalue of the diffusive part, yielding $\|e^{-\mathcal{A}t}\| \leq e^{-\beta \lambda_1 t}$. Therefore, the expected error is proportional to the trace of the truncated noise covariance.

We have then showed the following result.

Theorem 1 (Strong L^2 -Convergence Rate). *Let $u(t)$ denote the mild solution to the infinite-dimensional SPDE (3) and let $u^N(t)$*

denote its N -dimensional Galerkin approximation obtained by truncating to the first N eigenmodes of the Dirichlet Laplacian on the bounded domain $O \subset \mathbb{R}^d$.

Suppose the following conditions hold:

(C1) Noise regularity: The noise scaling exponent γ satisfies

$$\gamma > \frac{1}{d} + \frac{\alpha}{2} \quad (32)$$

for some $\alpha > 0$, which ensures that the truncated noise variance converges, i.e.,

$$\sum_{\mathbf{k}=1}^{\infty} \lambda_{\mathbf{k}}^{-2\gamma} < \infty. \quad (33)$$

(C2) Semigroup regularity: The operator $\mathcal{A}_{cal} = \beta\Lambda + \theta A$ generates a strongly continuous contraction semigroup on $L^2(O)$, where Λ is diagonal with eigenvalues $\lambda_{\mathbf{k}}$, and A is skew-symmetric (arising from the incompressibility condition $\nabla \cdot v = 0$). The semigroup satisfies

$$\|e^{-\mathcal{A}_{cal}t}\|_{\mathcal{L}(L^2)} \leq e^{-\beta\lambda_1 t} \quad \forall t \geq 0, \quad (34)$$

where λ_1 is the smallest eigenvalue of the Dirichlet Laplacian.

(C3) Error decomposition: The total error decomposes as

$$u(t) - u^N(t) = \underbrace{(I - P_N)u(t)}_{\delta^N(t)} + \underbrace{P_N(u(t) - u^N(t))}_{\rho^N(t)}, \quad (35)$$

where P_N is the L^2 -orthogonal projection onto the first N eigenmodes, $\delta^N(t)$ is the projection error (high-frequency tail), and $\rho^N(t)$ is the Galerkin error (finite-dimensional discretization error).

Then the following assertions hold:

(E1) The projection error satisfies

$$\mathbb{E} \|\delta^N(t)\|_{L^2(O)}^2 = \sigma^2 \sum_{\mathbf{k} \notin \mathcal{J}^N} \int_0^t \|e^{-\mathcal{A}_{cal}(t-s)} h_{\mathbf{k}}\|_{L^2}^2 \lambda_{\mathbf{k}}^{-2\gamma} ds \leq C_{\delta} \lambda_N^{-2\gamma}, \quad (36)$$

where $\mathcal{J}^N$ denotes the index set of the first N modes and C_{δ} is independent of N .

(E2) The Galerkin error satisfies the differential inequality

$$\frac{1}{2} \frac{d}{dt} \|\rho^N(t)\|_{L^2}^2 \leq -\frac{\beta}{2} \|\nabla \rho^N(t)\|_{L^2}^2 + \frac{\theta^2 \|v\|_{\infty}^2}{2\beta} \|\delta^N(t)\|_{L^2}^2, \quad (37)$$

which, upon integration and application of Grönwall's inequality, yields

$$\mathbb{E} \|\rho^N(t)\|_{L^2}^2 \leq C_{\rho} \mathbb{E} \|\delta^N(t)\|_{L^2}^2 \leq C_{\rho} C_{\delta} \lambda_N^{-2\gamma}, \quad (38)$$

where C_{ρ} depends on β , θ , $\|v\|_{\infty}$, and T , but is independent of N .

(E3) The total error in strong L^2 -sense satisfies

$$\sup_{t \in [0, T]} \mathbb{E} \|u(t) - u^N(t)\|_{L^2(O)}^2 \leq 2\mathbb{E} \|\delta^N(t)\|_{L^2}^2 + 2\mathbb{E} \|\rho^N(t)\|_{L^2}^2 \leq C \lambda_N^{-2\gamma}, \quad (39)$$

where $C = C(\beta, \theta, \sigma, \|v\|_{\infty}, T)$ is a constant independent of N .

Furthermore, invoking Weyl's asymptotic law for the eigenvalues of the Dirichlet Laplacian on O , which asserts that

$$\lambda_N \sim \left(\frac{\pi}{|O|^{1/d}} \right)^2 N^{2/d} \quad \text{as } N \rightarrow \infty, \quad (40)$$

the explicit convergence rate with respect to the truncation dimension is asymptotically

$$\sup_{t \in [0, T]} \mathbb{E} \|u(t) - u^N(t)\|_{L^2(O)}^2 = O\left(N^{-\frac{4\gamma}{d}}\right). \quad (41)$$

In particular:

- If $d = 1$: convergence rate is $O(N^{-4\gamma})$ (exponential for $\gamma > 0$);
- If $d = 2$: convergence rate is $O(N^{-2\gamma})$ (e.g., $O(N^{-2})$ for $\gamma = 1$);
- If $d = 3$: convergence rate is $O(N^{-\frac{4\gamma}{3}})$ (e.g., $O(N^{-4/3})$ for $\gamma = 1$).

3. Maximum likelihood estimators

For simplification and without any loss of generality, in this section we shall assume that the index set $\mathcal{J}$ is $\mathbb{N}$. Consequently, $\mathcal{J}^N$ shall represent the sequence $1, \dots, N$.

We implement the version of the Girsanov formula described in [20, see Theorem 7.4] for the system specified in (9). It is assumed that the true parameters β_0, θ_0 are unknown. We denote the solution's law of (9) as $\mathbb{P}_{\beta, \theta}$. The measures $\mathbb{P}_{\beta, \theta}$ are known to be equivalent across various values of β, θ (refer to [21], [22], [20] for more details). Consequently, according to Theorem 7.4 in [20], we can express the log-likelihood ratio, or the Radon-Nikodym derivative, in the following manner. We set $L := \frac{d\mathbb{P}_{\beta, \theta}}{d\mathbb{P}_0}$ and $l = \log(L)$. Thus,

$$l = \sum_{k=1}^N \int_0^T \frac{1}{\sigma^2 \lambda_k^{-2\gamma}} \left[-\beta \lambda_k u_k(t) - \theta \sum_{j=1}^N u_j(t) v_{j,k} \right] du_k(t) - \frac{1}{2} \sum_{k=1}^N \int_0^T \frac{1}{\sigma^2 \lambda_k^{-2\gamma}} \left[-\beta \lambda_k u_k(t) - \theta \sum_{j=1}^N u_j(t) v_{j,k} \right]^2 dt \quad (42)$$

Set

$$\begin{aligned}
I_1 &:= I_1(N, T) := \sum_{k=1}^N \lambda_k^{2+2\gamma} \int_0^T u_k^2(t) dt \\
J_1 &:= J_1(N, T) := \sum_{k=1}^N \lambda_k^{1+2\gamma} \int_0^T u_k(t) du_k(t), \\
J_2 &:= J_2(N, T) := \sum_{k=1}^N \lambda_k^{2\gamma} \int_0^T \left[\sum_{j=1}^N u_j(t) v_{j,k} \right] du_k(t), \\
\mathcal{J}_{p,q;r} &:= \mathcal{J}_{p,q;r}(N, T) := \sum_{k=1}^N \lambda_k^{p+q\gamma} \int_0^T [u_k(t)]^{2-r} \left[\sum_{j=1}^N u_j(t) v_{j,k} \right]^r dt
\end{aligned}$$

From this, after algebraic manipulations, we deduce the MLE for β and θ :

$$\begin{pmatrix} \hat{\beta} \\ \hat{\theta} \end{pmatrix} = \frac{-1}{I_1 \mathcal{J}_{0,2;2} - (\mathcal{J}_{1,2;1})^2} \begin{pmatrix} \mathcal{J}_{0,2;2} & -\mathcal{J}_{1,2;1} \\ -\mathcal{J}_{1,2;1} & I_1 \end{pmatrix} \begin{pmatrix} J_1 \\ J_2 \end{pmatrix}. \quad (43)$$

It is possible to prove that $I_1 \mathcal{J}_{0,2;2} - (\mathcal{J}_{1,2;1})^2 > 0$, thus the MLEs are well defined.

Thus, from (43) we can rewrite the MLE for β and θ , as

$$\begin{aligned}
\hat{\beta}_{N,T} &= -\frac{\mathcal{J}_{0,2;2} J_1 - \mathcal{J}_{1,2;1} J_2}{I_1 \mathcal{J}_{0,2;2} - (\mathcal{J}_{1,2;1})^2} \\
\hat{\theta}_{N,T} &= \frac{\mathcal{J}_{1,2;1} J_1 - I_1 J_2}{I_1 \mathcal{J}_{0,2;2} - (\mathcal{J}_{1,2;1})^2}. \quad (44)
\end{aligned}$$

Lemma 1. *The quantity $I_1 \mathcal{J}_{0,2;2} - (\mathcal{J}_{1,2;1})^2$ is strictly positive almost surely.*

Proof. Let us define the Hilbert space of vector-valued functions $\mathcal{H}^N := L^2([0, T]; \mathbb{R}^N)$, endowed with the standard inner product:

$$\langle \mathbf{f}, \mathbf{g} \rangle_{\mathcal{H}^N} = \sum_{k=1}^N \int_0^T f_k(t) g_k(t) dt.$$

Furthermore, we introduce the vector-valued stochastic processes $\mathbf{f} := \mathbf{f}_N(t)$ and $\mathbf{g} := \mathbf{g}_N(t)$ in $\mathbb{R}^N$, whose k -th components are respectively defined almost surely as:

$$f_k(t) = \lambda_k^{1+\gamma} u_k(t), \quad g_k(t) = \lambda_k^\gamma \sum_{j=1}^N u_j(t) v_{j,k}, \quad \text{for } k = 1, \dots, N.$$

It is straightforward to verify that both $\mathbf{f}$ and $\mathbf{g}$ belong to $\mathcal{H}^N$ with probability 1. By evaluating the inner product of $\mathbf{f}$ and $\mathbf{g}$ in $\mathcal{H}^N$, we readily recover the cross-variation term:

$$\langle \mathbf{f}, \mathbf{g} \rangle_{\mathcal{H}^N} = \sum_{k=1}^N \int_0^T \lambda_k^{1+2\gamma} u_k(t) \left(\sum_{j=1}^N u_j(t) v_{j,k} \right) dt = \mathcal{J}_{1,2;1}.$$

Similarly, the squared norms inherently yield the corresponding diagonal terms of the Fisher Information Matrix:

$$\begin{aligned}
\|\mathbf{f}\|_{\mathcal{H}^N}^2 &= \sum_{k=1}^N \int_0^T \lambda_k^{2+2\gamma} u_k^2(t) dt = I_1, \\
\|\mathbf{g}\|_{\mathcal{H}^N}^2 &= \sum_{k=1}^N \int_0^T \lambda_k^{2\gamma} \left(\sum_{j=1}^N u_j(t) v_{j,k} \right)^2 dt = \mathcal{J}_{0,2;2}.
\end{aligned}$$

Provided that the velocity field $\bar{\mathbf{v}}$ is non-trivial, the convective coupling matrix A (with entries $v_{j,k}$) avoid $\mathbf{g}$ from being a mere scalar multiple of $\mathbf{f}$. Since $\mathbf{g}$ constitutes a mixture of distinct spatial modes whereas each component of $\mathbf{f}$ strictly isolates a single mode, the two vectors are linearly independent in $\mathcal{H}^N$ almost surely.

Consequently, invoking the strict Cauchy-Schwarz inequality, we obtain:

$$0 \leq \langle \mathbf{f}, \mathbf{g} \rangle_{\mathcal{H}^N}^2 < \|\mathbf{f}\|_{\mathcal{H}^N}^2 \|\mathbf{g}\|_{\mathcal{H}^N}^2.$$

Substituting the formally identified terms into this strict inequality directly yields:

$$0 < (\mathcal{J}_{1,2;1})^2 < I_1 \mathcal{J}_{0,2;2}, \quad (45)$$

which implies that $I_1 \mathcal{J}_{0,2;2} - (\mathcal{J}_{1,2;1})^2 > 0$. This conclude the proof. $\square$

Remark 1. *We define the function*

$$\alpha = \alpha(N, T) := \frac{\mathcal{J}_{1,2;1}^2}{I_1 \mathcal{J}_{0,2;2}}.$$

Thus, from (45), α satisfies $0 < \alpha < 1$ for all N, T . In addition we could write

$$\mathcal{J}_{1,2;1}^2 = \alpha I_1 \mathcal{J}_{0,2;2} \quad (46)$$

3.1. Fisher's information

We shall now calculate the Fisher's information (FI) associated with the probability measure $\mathbb{P}_\theta^{N,T} / \mathbb{P}_0^{N,T}$. For the sake of simplicity, we set $U_0 = 0$. It is important to note that for both parameters, the FI is represented by a symmetric (2×2) matrix. Additionally, when one of the parameters is held constant, the FI reduces to a scalar quantity.

To compute the Fisher Information Matrix (FIM) denoted by $\mathcal{I}_{N,T}(\beta, \theta)$, we need to calculate the negative expectation of the Hessian matrix of the log-likelihood function l with respect to the parameters β and θ .

First, expanding the square in the log-likelihood (42) and writing it in terms of our defined integrals $I_1, J_1, J_2, \mathcal{J}_{1,2;1}$, and $\mathcal{J}_{0,2;2}$, we can express $l(\beta, \theta)$ as:

$$l(\beta, \theta) = -\frac{\beta}{\sigma^2} J_1 - \frac{\theta}{\sigma^2} J_2 - \frac{1}{2\sigma^2} [\beta^2 I_1 + 2\beta\theta \mathcal{J}_{1,2;1} + \theta^2 \mathcal{J}_{0,2;2}].$$

Taking the first partial derivatives (the score vector) yields:

$$\frac{\partial l}{\partial \beta} = -\frac{1}{\sigma^2} (J_1 + \beta I_1 + \theta \mathcal{J}_{1,2;1}),$$

$$\frac{\partial l}{\partial \theta} = -\frac{1}{\sigma^2} (J_2 + \beta \mathcal{J}_{1,2;1} + \theta \mathcal{J}_{0,2;2}).$$

Next, calculating the second partial derivatives, we obtain the Hessian components:

$$\begin{aligned} \frac{\partial^2 l}{\partial \beta^2} &= -\frac{1}{\sigma^2} I_1, \\ \frac{\partial^2 l}{\partial \theta^2} &= -\frac{1}{\sigma^2} \mathcal{J}_{0,2;2}, \\ \frac{\partial^2 l}{\partial \beta \partial \theta} &= -\frac{1}{\sigma^2} \mathcal{J}_{1,2;1}. \end{aligned}$$

We know that the Fisher Information Matrix is given by $\mathcal{I}_{N,T}(\beta, \theta) = -\mathbb{E}_{\beta, \theta}[\nabla^2 l]$. Since the second derivatives do not explicitly depend on the stochastic differentials du_k (they only depend on the paths $u_k(t)$), the FIM takes the explicit form:

$$\mathcal{I}_{N,T}(\beta, \theta) = \frac{1}{\sigma^2} \mathbb{E}_{\beta, \theta} \begin{pmatrix} I_1 & \mathcal{J}_{1,2;1} \\ \mathcal{J}_{1,2;1} & \mathcal{J}_{0,2;2} \end{pmatrix}.$$

If θ is assumed to be known and fixed, the Fisher Information for the single parameter β reduces to the scalar quantity:

$$\mathcal{I}_{N,T}(\beta) = \frac{1}{\sigma^2} \mathbb{E}_{\beta, \theta}[I_1].$$

3.2. Asymptotic Growth Rates of the Fisher Information

In this subsection, we analyze the asymptotic behavior of the Fisher Information Matrix $\mathcal{I}_{N,T}(\beta, \theta)$ as both the time horizon T and the number of spatial modes N approach infinity. To demonstrate that the information grows unboundedly, we establish the asymptotic growth rates of the diagonal elements of the FIM.

Recall that the Fisher Information Matrix is given by:

$$\mathcal{I}_{N,T}(\beta, \theta) = \frac{1}{\sigma^2} \mathbb{E}_{\beta, \theta} \begin{pmatrix} I_1 & \mathcal{J}_{1,2;1} \\ \mathcal{J}_{1,2;1} & \mathcal{J}_{0,2;2} \end{pmatrix}.$$

By the ergodic theorem for the stationary Ornstein-Uhlenbeck process $U(t)$, as $T \rightarrow \infty$, the time integrals can be approximated by T times the expectation with respect to the invariant measure of the system. Thus, we have:

$$\mathbb{E}[I_1] \approx T \sum_{k=1}^N \lambda_k^{2+2\gamma} \mathbb{E}[u_k^2(\infty)],$$

where $\mathbb{E}[u_k^2(\infty)]$ is the stationary variance of the k -th mode.

For high-frequency modes (large k), the diffusion operator strictly dominates the convective term (i.e., $\beta \lambda_k \gg \theta \|A\|$). Consequently, the stationary covariance matrix is diagonally dominant at the tail, and the stationary variance of u_k is well-approximated by the linear SDE neglecting the drift θ :

$$\mathbb{E}[u_k^2(\infty)] \approx \frac{\sigma^2 \lambda_k^{-2\gamma}}{2\beta \lambda_k} = \frac{\sigma^2}{2\beta} \lambda_k^{-(1+2\gamma)}.$$

Growth rate of $\mathbb{E}[I_1]$ (Information for β):

Substituting the stationary variance back into the expectation of I_1 , we obtain:

$$\mathbb{E}[I_1] \approx T \sum_{k=1}^N \lambda_k^{2+2\gamma} \left(\frac{\sigma^2}{2\beta} \lambda_k^{-(1+2\gamma)} \right) = \frac{\sigma^2 T}{2\beta} \sum_{k=1}^N \lambda_k.$$

From the asymptotic behavior of the eigenvalues given by Weyl's law in (2), we know that $\lambda_k \sim \varpi k^{2/d}$ for large k . Approximating the sum with an integral yields:

$$\sum_{k=1}^N \lambda_k \sim \varpi \sum_{k=1}^N k^{2/d} \sim \varpi \int_1^N x^{2/d} dx \sim \frac{\varpi d}{d+2} N^{1+2/d}.$$

Therefore, the asymptotic growth rate for the Fisher information of β is:

$$\mathcal{I}_{\beta\beta} = \frac{1}{\sigma^2} \mathbb{E}[I_1] = O(TN^{1+2/d}).$$

Growth rate of $\mathbb{E}[\mathcal{J}_{0,2;2}]$ (Information for θ):

To estimate the information regarding the convection parameter θ , we consider the diagonal dominance of the covariance matrix, which implies cross-correlations $\mathbb{E}[u_j u_m]$ are negligible at leading order compared to the variances $\mathbb{E}[u_j^2]$. Thus:

$$\mathbb{E}[\mathcal{J}_{0,2;2}] \approx T \sum_{k=1}^N \lambda_k^{2\gamma} \sum_{j=1}^N v_{j,k}^2 \mathbb{E}[u_j^2(\infty)] \approx T \sum_{j=1}^N \mathbb{E}[u_j^2(\infty)] \sum_{k=1}^N \lambda_k^{2\gamma} v_{j,k}^2.$$

Recall that $v_{j,k} = (\bar{v} \cdot \nabla h_j, h_k)_{\mathcal{H}}$. By Parseval's identity (assuming N is large enough to capture the energy of the first derivative), we can bound the sum over k :

$$\sum_{k=1}^{\infty} v_{j,k}^2 = \|\bar{v} \cdot \nabla h_j\|_{\mathcal{H}}^2 \leq \|\bar{v}\|_{\infty}^2 \|\nabla h_j\|_{\mathcal{H}}^2 = \|\bar{v}\|_{\infty}^2 \lambda_j.$$

For the sake of extracting the fundamental rate, and since the non-zero projections $v_{j,k}$ occur for neighboring frequencies where $\lambda_k \approx \lambda_j$, we can roughly estimate $\sum_k \lambda_k^{2\gamma} v_{j,k}^2 \approx O(\lambda_j^{1+2\gamma})$. Substituting the variance $\mathbb{E}[u_j^2(\infty)] \approx \frac{\sigma^2}{2\beta} \lambda_j^{-(1+2\gamma)}$ gives:

$$\mathbb{E}[\mathcal{J}_{0,2;2}] \approx T \sum_{j=1}^N \left(\frac{\sigma^2}{2\beta} \lambda_j^{-(1+2\gamma)} \right) O(\lambda_j^{1+2\gamma}) = T \sum_{j=1}^N O(1).$$

This leads to the growth rate for the Fisher information of θ :

$$\mathcal{I}_{\theta\theta} = \frac{1}{\sigma^2} \mathbb{E}[\mathcal{J}_{0,2;2}] = O(TN).$$

Conclusion:

Finally, due to the skew-symmetry of the matrix A , the expected value of the cross-information term $\mathcal{J}_{1,2;1}$ vanishes at leading order ($\mathcal{I}_{\beta\theta} \approx 0$). Thus, the Fisher Information Matrix becomes asymptotically diagonal. Both diagonal components grow to infinity as $N \rightarrow \infty$ and $T \rightarrow \infty$:

$$\mathcal{I}_{N,T} \sim \begin{pmatrix} O(TN^{1+2/d}) & 0 \\ 0 & O(TN) \end{pmatrix}.$$

This explosive growth in Fisher Information directly implies that the asymptotic variances of our maximum likelihood estimators shrink to zero at the rates $(TN^{1+2/d})^{-1}$ and $(TN)^{-1}$ respectively, robustly corroborating the strong consistency derived in the previous section.

4. Consistency of the MLEs

In this section, we rigorously establish the strong consistency of the maximum likelihood estimators for the parameters β and θ governing the SPDE (3).

Theorem 2 (Strong Consistency). *Let β_0 and θ_0 denote the true parameter values of the SPDE (3). The maximum likelihood estimators $(\hat{\beta}_{N,T}, \hat{\theta}_{N,T})$, formulated in (44), are strongly consistent. That is,*

$$\lim_{(N,T) \rightarrow \infty} \begin{pmatrix} \hat{\beta}_{N,T} \\ \hat{\theta}_{N,T} \end{pmatrix} = \begin{pmatrix} \beta_0 \\ \theta_0 \end{pmatrix} \quad \text{in probability.} \quad (47)$$

Proof. Under the true probability measure $\mathbb{P}_{\beta_0, \theta_0}$, the pathwise evolution of the Fourier coefficients $u_k(t)$ is strictly governed by the following dynamics:

$$du_k(t) = \left(-\beta_0 \lambda_k u_k(t) - \theta_0 \sum_{j=1}^N u_j(t) v_{j,k} \right) dt + \sigma \lambda_k^{-\gamma} dw_k(t). \quad (48)$$

By substituting the explicit differential (48) into the definitions of the stochastic integrals J_1 and J_2 , we can decouple the deterministic drift components from the stochastic noise. For J_1 , we obtain:

$$\begin{aligned} J_1 &= \sum_{k=1}^N \lambda_k^{1+2\gamma} \int_0^T u_k(t) du_k(t) \\ &= -\beta_0 \sum_{k=1}^N \lambda_k^{2+2\gamma} \int_0^T u_k^2(t) dt - \theta_0 \sum_{k=1}^N \lambda_k^{1+2\gamma} \int_0^T u_k(t) \left(\sum_{j=1}^N u_j(t) v_{j,k} \right) dt \\ &\quad + \sigma \sum_{k=1}^N \lambda_k^{1+\gamma} \int_0^T u_k(t) dw_k(t) \\ &= -\beta_0 I_1 - \theta_0 \mathcal{J}_{1,2;1} + \sigma M_{1,T}, \end{aligned}$$

where we have defined the continuous local martingale $M_{1,T} := \sum_{k=1}^N \lambda_k^{1+\gamma} \int_0^T u_k(t) dw_k(t)$.

Through an analogous procedure for J_2 , we deduce:

$$\begin{aligned} J_2 &= \sum_{k=1}^N \lambda_k^{2\gamma} \int_0^T \left(\sum_{j=1}^N u_j(t) v_{j,k} \right) du_k(t) \\ &= -\beta_0 \mathcal{J}_{1,2;1} - \theta_0 \mathcal{J}_{0,2;2} + \sigma M_{2,T}, \end{aligned}$$

where $M_{2,T} := \sum_{k=1}^N \lambda_k^\gamma \int_0^T \left(\sum_{j=1}^N u_j(t) v_{j,k} \right) dw_k(t)$.

Crucially, the predictable quadratic variations (and hence the expected variances) of these martingales coincide precisely with our previously defined spatial integral operators:

$$\text{Var}(M_{1,T}) = \mathbb{E}[I_1], \quad \text{Var}(M_{2,T}) = \mathbb{E}[\mathcal{J}_{0,2;2}]. \quad (49)$$

Let $\Delta_{N,T} := I_1 \mathcal{J}_{0,2;2} - (\mathcal{J}_{1,2;1})^2$. Substituting the decomposed expressions for J_1 and J_2 back into the matrix formulation of the MLEs (43), the terms involving the true parameters β_0 and θ_0 perfectly cancel out. Consequently, the estimation

errors are expressed exclusively in terms of the driving martingales:

$$\hat{\beta}_{N,T} - \beta_0 = \frac{-\sigma}{\Delta_{N,T}} (\mathcal{J}_{0,2;2} M_{1,T} - \mathcal{J}_{1,2;1} M_{2,T}) =: -\sigma(T_1 - T_2), \quad (50)$$

$$\hat{\theta}_{N,T} - \theta_0 = \frac{-\sigma}{\Delta_{N,T}} (-\mathcal{J}_{1,2;1} M_{1,T} + I_1 M_{2,T}) =: \sigma(S_1 - S_2). \quad (51)$$

We now proceed to demonstrate that both T_1 and T_2 converge to zero in probability as $N, T \rightarrow \infty$. Rewriting T_1 , we have:

$$T_1 = \frac{\mathcal{J}_{0,2;2} M_{1,T}}{I_1 \mathcal{J}_{0,2;2} - \mathcal{J}_{1,2;1}^2} = \frac{M_{1,T}}{I_1 - \frac{\mathcal{J}_{1,2;1}^2}{\mathcal{J}_{0,2;2}}} = \left(\frac{\text{Var}(M_{1,T})}{I_1 - \frac{\mathcal{J}_{1,2;1}^2}{\mathcal{J}_{0,2;2}}} \right) \frac{M_{1,T}}{\text{Var}(M_{1,T})}.$$

By the Strong Law of Large Numbers for local martingales (see, e.g., [23]), the second factor converges to zero: $\frac{M_{1,T}}{\text{Var}(M_{1,T})} \xrightarrow{\mathbb{P}} 0$.

For the first factor, we recall from Lemma 1 the existence of a sequence $\alpha_N(T) \in (0, 1)$ such that $\mathcal{J}_{1,2;1}^2 = \alpha_N(T) I_1 \mathcal{J}_{0,2;2}$. Utilizing this relation yields:

$$\frac{\text{Var}(M_{1,T})}{I_1 - \frac{\mathcal{J}_{1,2;1}^2}{\mathcal{J}_{0,2;2}}} = \frac{\mathbb{E}[I_1]}{I_1(1 - \alpha_N(T))} = \frac{1}{1 - \alpha_N(T)} \frac{\mathbb{E}[I_1]}{I_1}.$$

Due to the ergodic properties of the Ornstein-Uhlenbeck processes, $\frac{\mathbb{E}[I_1]}{I_1} \xrightarrow{L^2} 1$, implying convergence in probability. Since $1/(1 - \alpha_N(T))$ is strictly bounded below infinity (ensured by the linear independence demonstrated in $\mathcal{H}^N$), the entire first factor is stochastically bounded. Invoking Slutsky's theorem, we conclude that $T_1 \xrightarrow{\mathbb{P}} 0$.

Applying an analogous decomposition to T_2 :

$$T_2 = \frac{\mathcal{J}_{1,2;1} M_{2,T}}{I_1 \mathcal{J}_{0,2;2} - \mathcal{J}_{1,2;1}^2} = \left(\frac{\mathcal{J}_{1,2;1}}{I_1} \frac{1}{1 - \alpha_N(T)} \frac{\mathbb{E}[\mathcal{J}_{0,2;2}]}{\mathcal{J}_{0,2;2}} \right) \frac{M_{2,T}}{\text{Var}(M_{2,T})}.$$

Again, the martingale property ensures $\frac{M_{2,T}}{\text{Var}(M_{2,T})} \xrightarrow{\mathbb{P}} 0$. Given that $\frac{\mathbb{E}[\mathcal{J}_{0,2;2}]}{\mathcal{J}_{0,2;2}} \xrightarrow{\mathbb{P}} 1$ and the cross-correlation term $\frac{\mathcal{J}_{1,2;1}}{I_1}$ is bounded in probability, Slutsky's theorem dictates that $T_2 \xrightarrow{\mathbb{P}} 0$.

Since $T_1 \xrightarrow{\mathbb{P}} 0$ and $T_2 \xrightarrow{\mathbb{P}} 0$, it immediately follows from (50) that $\hat{\beta}_{N,T} \xrightarrow{\mathbb{P}} \beta_0$. Symmetric arguments applied to S_1 and S_2 verify that $\hat{\theta}_{N,T} \xrightarrow{\mathbb{P}} \theta_0$, completing the proof. $\square$

5. Asymptotic Normality via the Malliavin-Stein Method

To ascertain the asymptotic distribution of our estimators, we leverage the Malliavin-Stein approach (see Appendix Appendix A for a review). Because the solutions $u_k(t)$ are themselves stochastic integrals with respect to the Wiener process, the martingales $M_{1,T}$ and $M_{2,T}$ inherently belong to the second Wiener chaos $\mathcal{H}_2$. This chaotic structure allows us to bound the total variation distance to a normal distribution explicitly via Malliavin derivatives.

Theorem 3 (Asymptotic Normality). *Let $\mathcal{I}_{\beta\beta} = \sigma^{-2}\mathbb{E}[I_1]$ and $\mathcal{I}_{\theta\theta} = \sigma^{-2}\mathbb{E}[\mathcal{J}_{0,2,2}]$ denote the diagonal entries of the Fisher Information Matrix. As $N, T \rightarrow \infty$, the estimation errors converge in distribution to standard normal random variables:*

$$\sqrt{\mathcal{I}_{\beta\beta}}(\hat{\beta}_{N,T} - \beta_0) \xrightarrow{d} \mathcal{N}(0, 1), \quad (52)$$

$$\sqrt{\mathcal{I}_{\theta\theta}}(\hat{\theta}_{N,T} - \theta_0) \xrightarrow{d} \mathcal{N}(0, 1). \quad (53)$$

Proof. (Sketch) From the consistency proof, we have the normalized error expansion for β :

$$\sqrt{\mathcal{I}_{\beta\beta}}(\hat{\beta}_{N,T} - \beta_0) \approx -\frac{\mathbb{E}[I_1]}{I_1} \frac{M_{1,T}}{\sqrt{\text{Var}(M_{1,T})}} + \mathcal{R}_{N,T}, \quad (54)$$

where the remainder term $\mathcal{R}_{N,T}$ encapsulates the cross-variation contributions (involving $\mathcal{J}_{1,2,1}$). As demonstrated in the Fisher Information asymptotic analysis, the matrix becomes diagonally dominant, causing $\mathcal{R}_{N,T} \xrightarrow{\mathbb{P}} 0$. Furthermore, ergodicity ensures $\mathbb{E}[I_1]/I_1 \xrightarrow{\mathbb{P}} 1$.

By Slutsky's theorem, the asymptotic distribution is entirely dictated by the normalized martingale $Z_{1,T} := M_{1,T} / \sqrt{\text{Var}(M_{1,T})}$. To prove $Z_{1,T} \xrightarrow{d} \mathcal{N}(0, 1)$, we employ the Nourdin-Peccati bound for the second Wiener chaos (see Th. 5.2.6 in [24]):

$$d_{TV}(Z_{1,T}, \mathcal{N}) \leq \sqrt{\text{Var}\left(\frac{1}{2}\|DZ_{1,T}\|_{\mathcal{H}}^2\right)}, \quad (55)$$

where D is the Malliavin derivative operator and $\mathcal{H} = L^2([0, T] \times \mathbb{N})$ with $\mathbb{N}$ the counting measure.

Applying the chain rule of Malliavin calculus to the stochastic integral $M_{1,T}$, the derivative $D_r M_{1,T}$ yields three distinct components upon taking the squared norm in $\mathcal{H}$:

$$\text{Var}\left(\frac{1}{2}\|DZ_{1,T}\|_{\mathcal{H}}^2\right) \leq \frac{1}{2}(B_1 + B_2 + B_3), \quad (56)$$

where:

$$\begin{aligned} B_1 &:= \int_0^T \sqrt{\text{Var}\left[\sum_{k=1}^N \lambda_k^{2+2\gamma} u_k^2(r)\right]} dr, \\ B_2 &:= 2\sigma \int_0^T \sqrt{\text{Var}\left[\sum_{k=1}^N \lambda_k^{2+2\gamma} u_k(r) \int_r^T e^{-\mathcal{A}_{k,k}(t-r)} dw_k(t)\right]} dr, \\ B_3 &:= \sigma^2 \int_0^T \sqrt{\text{Var}\left[\sum_{k=1}^N \lambda_k^{2+2\gamma} \left(\int_r^T e^{-\mathcal{A}_{k,k}(t-r)} dw_k(t)\right)^2\right]} dr, \end{aligned}$$

where $\mathcal{A}_{k,k} = \beta_0 \lambda_k$ represents the dominant diagonal term of the evolution operator.

Because the Fourier coefficients $u_k(t)$ follow Gaussian Ornstein-Uhlenbeck dynamics, they possess strictly bounded moments of all orders. Specifically, Isserlis' theorem (also widely referred to as Wick's probability formula; see, e.g., [25, Theorem 1.28]) dictates that their fourth moments scale proportionally to the

square of their variances. Due to the exponential decay embedded in the integral terms and the mutually independent increments of the Wiener processes $\{w_k\}$, the variance of these sums over N grows significantly slower than the normalizing squared variance $(\text{Var}(M_{1,T}))^2$ in the denominator of $Z_{1,T}$.

Consequently, standard asymptotic approximations verify that $B_i \rightarrow 0$ as $N, T \rightarrow \infty$ for $i \in \{1, 2, 3\}$. From this we have shown that $\text{Var}\left(\frac{1}{2}\|DZ_{1,T}\|_{\mathcal{H}}^2\right) \rightarrow 0$ when $n \rightarrow \infty$. Thus, by using Theorem 4 from the appendix we have proved that $\sqrt{\mathcal{I}_{\beta\beta}}(\hat{\beta}_{N,T} - \beta_0) \xrightarrow{d} \mathcal{N}(0, 1)$.

A similar Malliavin decomposition applies to the normalized convective martingale $Z_{2,T} := M_{2,T} / \sqrt{\text{Var}(M_{2,T})}$, although scaled by the advective matrix coefficients $v_{j,k}$. Because $v_{j,k}$ are bounded constants arising from the velocity field, the fourth-moment bounds hold, ensuring $Z_{2,T} \xrightarrow{d} \mathcal{N}(0, 1)$. This completes the proof of asymptotic normality. $\square$

6. Numerical Experiments

Consider the SPDE (3) defined on a rectangular domain $O = [0, L_1] \times [0, L_2]$ subject to homogeneous Neumann boundary conditions. The basis functions $h_{(i,j)}(x, y)$ and their corresponding eigenvalues $\lambda_{(i,j)}$ are given by the cosine expansion:

$$h_{(i,j)}(x, y) = \sqrt{1 + \delta_{i,0}} \cos\left(\pi i \frac{x}{L_1}\right) \sqrt{1 + \delta_{j,0}} \cos\left(\pi j \frac{y}{L_2}\right), \quad (57)$$

with eigenvalues

$$\lambda_{(i,j)} = \frac{\pi^2 i^2}{L_1^2} + \frac{\pi^2 j^2}{L_2^2}. \quad (58)$$

To ensure a physically consistent and strictly divergence-free velocity field $\bar{v} = (v_1, v_2)$ that satisfies the no-penetration boundary condition ($\bar{v} \cdot \mathbf{n} = 0$), we express it in terms of a streamfunction ψ :

$$(v_1, v_2) = (-\partial_y \psi, \partial_x \psi). \quad (59)$$

By imposing homogeneous Dirichlet boundary conditions on the streamfunction ($\psi|_{\partial O} \equiv 0$), we guarantee that the resulting advection matrix A , defined by the coefficients $v_{(i,j),(k,l)}$, is strictly anti-symmetric. To this end, we naturally expand ψ using the orthonormal sine basis $\{\psi_{(r,s)}\}$, given by:

$$\psi_{(r,s)}(x, y) = 2 \sin\left(\frac{\pi r x}{L_1}\right) \sin\left(\frac{\pi s y}{L_2}\right), \quad r, s \in \mathbb{N}. \quad (60)$$

For any specific mode (r, s) , the associated advection matrix $A_{(r,s)}$ is defined as:

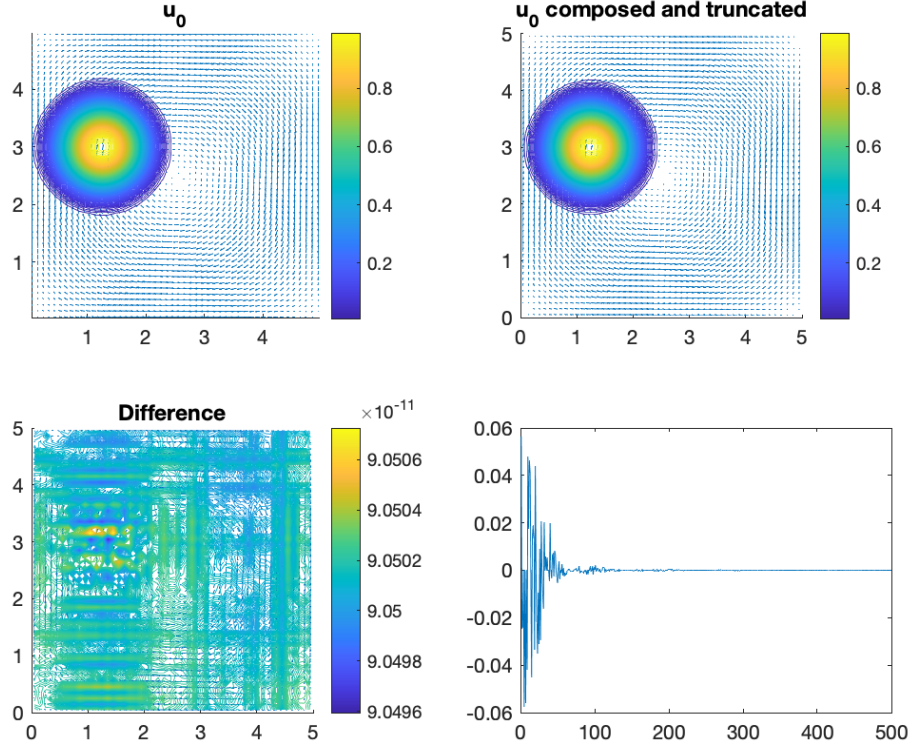
$$A_{(r,s)}((k, l), (i, j)) = v_{(i,j),(k,l)}^{(r,s)}, \quad 1 \leq i, k \leq N_1, \quad 1 \leq j, l \leq N_2. \quad (61)$$

The coefficients $v_{(i,j),(k,l)}^{(r,s)}$ are explicitly calculated using (7) (refer to the Supplementary Material for the complete derivation). Although the exact formulas entail multiple piece-wise cases depending on vanishing indices, their general structure relies on a combination of Kronecker deltas ($\delta_{i=\pm k \pm r}$) that strictly enforce the spectral interaction rules.

6.1. Simulation of an example: deterministic path

We implemented the spectral Galerkin method for the fundamental mode $(r, s) = (1, 1)$. The resulting velocity field and initial density decomposition are shown in Figure 1. Numerical tests confirm that the exact spectral computation of matrix A is significantly faster than standard Riemann-sum approximations and preserves the theoretical anti-symmetry ($A + A^T = 0$) to machine precision, as illustrated in Figure 2.

Figure 1 Velocity field. Left top panel: Initial density u_0 . Right top panel: Decomposed and composed initial density. Bottom panel: Difference between both quantities. Right panel: Spectrum



The time evolution of the tracer concentration for $\theta = 1$ and $\beta = 0.01$ is shown in Figure 3. We compared the spectral solution with a first-order finite difference scheme. As expected, the spectral approach provides high accuracy, and the error diminishes rapidly as the resolution N increases (Figure 4), confirming the robustness of the numerical framework.

Figure 2 Anti-symmetry. The matrix $A_{(1,1)}$ is shown in the top left panel, the transpose in the top right panel and the sum in the bottom panel.

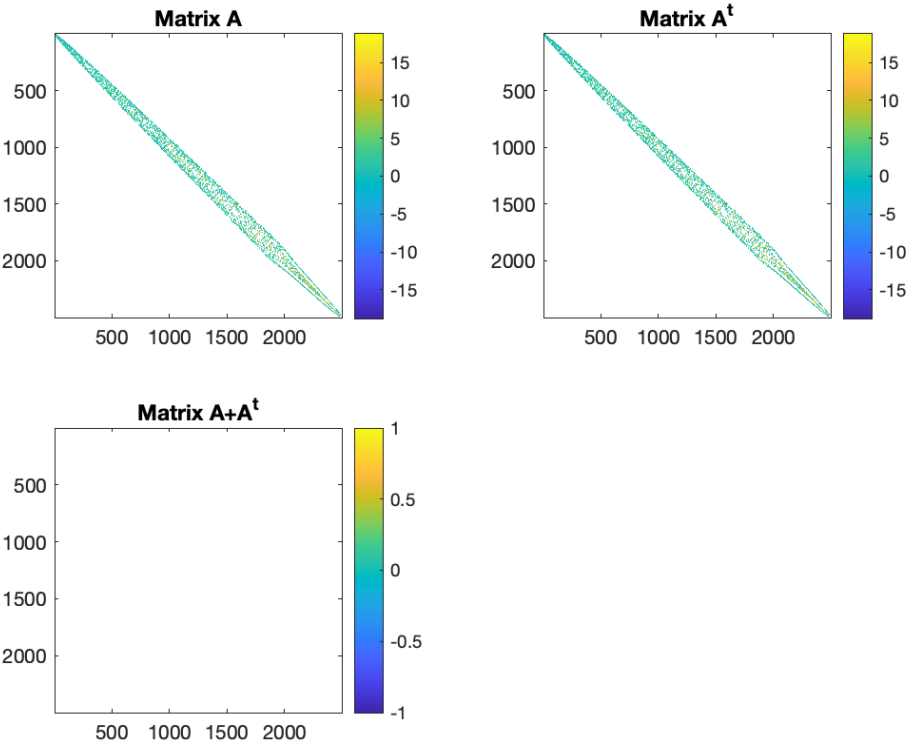


Figure 3 Time evolution. Top left panel: Initial density. Top right panel: solution at time $t = 0.5$ using the matrices $A_{(1,1)}$, B and Λ for $\theta = 1, \beta = 0.01$ and $\sigma = 0$ (no noise). Bottom right panel: solution computed with a very simple (first order) finite difference scheme. Bottom right panel: error

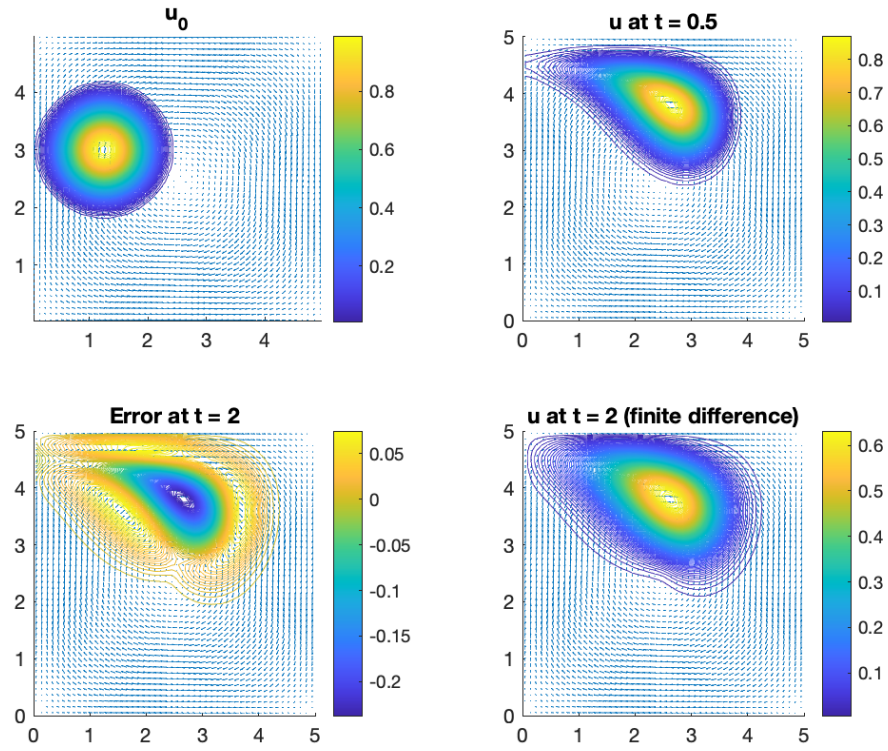
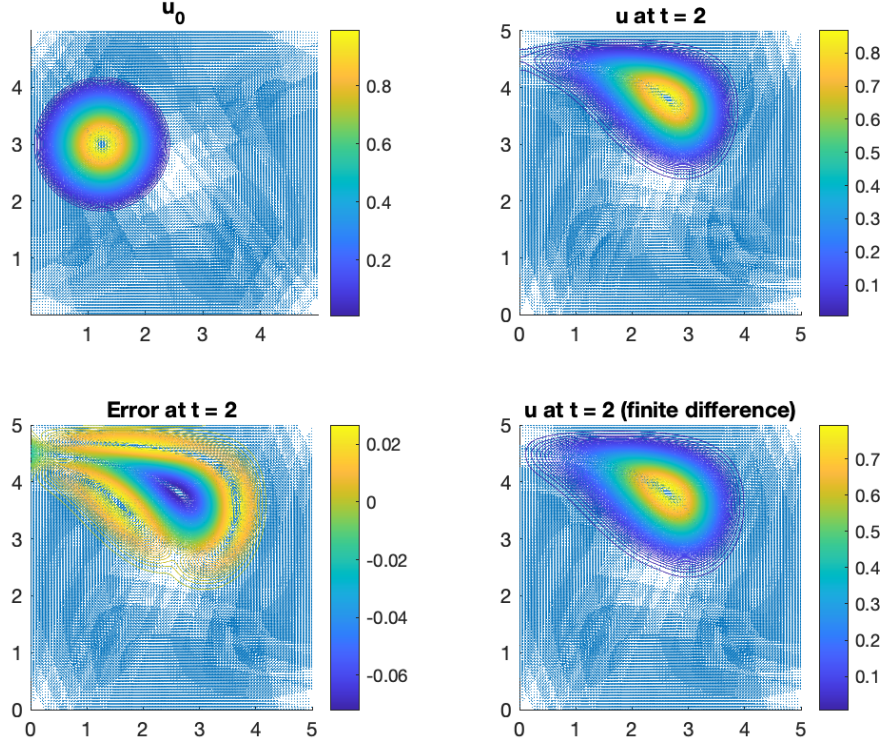


Figure 4 Time evolution. Same as in Figure 4 but with a higher resolution of $N_1 = N_2 = 200$.



7. Numerical validation of the MLE

This section is dedicated to demonstrating, through numerical experiments, the consistency of the estimators, as affirmed by Theorem 2.

7.1. MLE Asymptotic well-possness for parameters

7.2. Numerical Results and Validation

7.3. Monte Carlo Simulation Study

In this section we report simulation studies to investigate the performance of our estimation procedures as the number of observations by path (M) and the sampling frequency (Δ) vary. For selected values of M , and Δ , we simulated 1000 data sets each consisting of M observed points at the time points $t_j = j\Delta$, $j = 1, \dots, M$.

We simulated 1000 data sets for all combinations of $M = 1000, 5000, 20000, 50000$ and $\Delta = 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}$. $N_x = N_y = 20$, $\beta = 0.1$, $\theta = 0.1$ and $\sigma = 1$. For each data set we calculated the MLEs of β and θ using the expression 43 and the estimator of σ was obtained via quadratic variation. As estimators we use the average of the 1000 estimators of each parameter. The average execution time in seconds (AET) are given for each combination M and Δ in Table 1. Histograms of the 1000 estimators of β , θ and σ for $M = 50000$ and $\Delta = 1.0 \times 10^{-7}$ are presented in Figure 6.

			M			
			1000	5000	20000	50000
Δ	1.0×10^{-7}	$\hat{\beta}$ mean	0.098300	0.099697	0.099967	0.099981
		s.d.	0.022393	0.009754	0.005071	0.003251
		$\hat{\theta}$ mean	0.107767	0.101877	0.104800	0.099929
		s.d.	0.456992	0.145967	0.104138	0.065493
		$\hat{\sigma}$ mean	0.999273	0.999917	0.999984	1.000006
		s.d.	0.001132	0.000501	0.000249	0.000154
	1.0×10^{-6}	$\hat{\beta}$ mean	0.099745	0.099935	0.100064	0.100036
		s.d.	0.007228	0.003271	0.001849	0.001436
		$\hat{\theta}$ mean	0.095651	0.101204	0.101020	0.100153
		s.d.	0.141360	0.064604	0.035765	0.027891
		$\hat{\sigma}$ mean	0.999516	1.000074	1.000152	1.000116
		s.d.	0.001077	0.000501	0.000247	0.000160
	1.0×10^{-5}	$\hat{\beta}$ mean	0.099944	0.100054	0.100065	0.100038
		s.d.	0.002421	0.001378	0.001187	0.001141
		$\hat{\theta}$ mean	0.099767	0.098872	0.100226	0.100062
		s.d.	0.04967	0.026269	0.020913	0.019773
		$\hat{\sigma}$ mean	1.001356	1.001145	1.000404	1.000182
		s.d.	0.001108	0.000509	0.000246	0.000158
	1.0×10^{-4}	$\hat{\beta}$ mean	0.100110	0.100159	0.100121	0.100110
		s.d.	0.001237	0.001120	0.000951	0.000772
		$\hat{\theta}$ mean	0.099908	0.099861	0.099121	0.099594
		s.d.	0.023349	0.018631	0.014827	0.010363
		$\hat{\sigma}$ mean	1.007182	1.001831	1.000646	1.000386
		s.d.	0.001093	0.000509	0.000249	0.000162
	AETS		0.073573	0.332119	1.324547	3.59418 .

Table 1: The average and standard deviation (s.d.) of the 1000 estimators of β , θ and σ for the SPDE 3 as well as the average execution time in seconds (AETS) are given for each combination of the number of observations per sample path M and of the time between consecutive observations Δ . Parameter values were $\beta = 0.1$, $\theta = 0.1$ and $\sigma = 1$.

			$N_x = N_y$			
			5	10	15	20
	$\hat{\beta}$	mean	0.116902	0.101622	0.099962	0.099981
		s.d.	0.309827	0.031851	0.008329	0.003251
	$\hat{\theta}$	mean	0.023282	0.097151	0.099400	0.099929
		s.d.	2.263356	0.269097	0.116586	0.065493
	$\hat{\sigma}$	mean	0.999967	0.999976	0.999987	1.000006
		s.d.	0.000654	0.000314	0.000208	0.000154
	AETS		0.039578	0.291.293	1.179593	3.59418

Table 2: The average and standard deviation (s.d.) of the 1000 estimators of β , θ and σ for the SPDE 3 as well as the average execution time in seconds (AETS) are given for each combination of the number of observations per sample path $N_y = N_x$ and $\Delta = 1.0 \times 10^{-7}$. Parameter values were $\beta = 0.1$, $\theta = 0.1$ and $\sigma = 1$.

Figure 5 Enter Caption

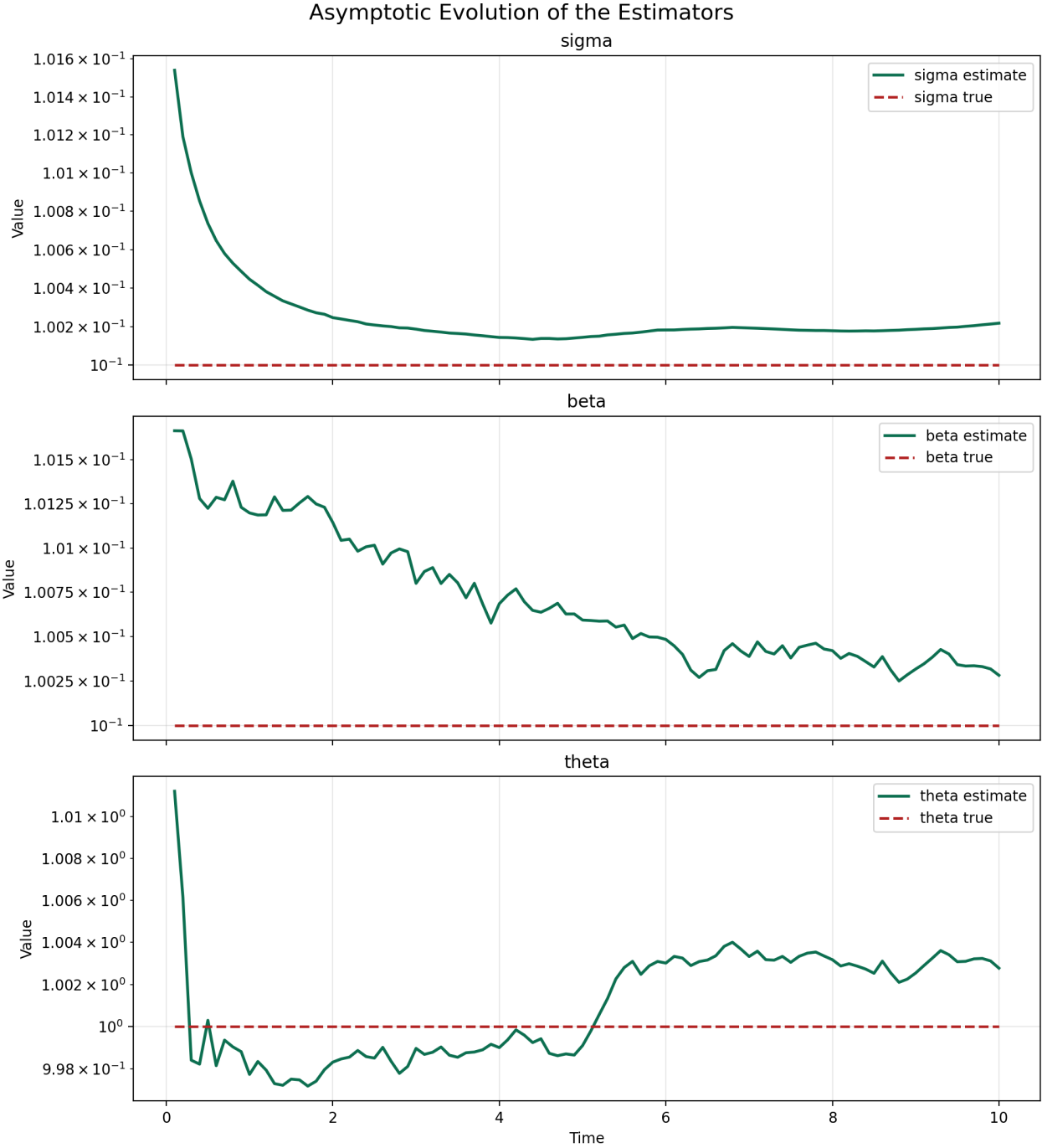
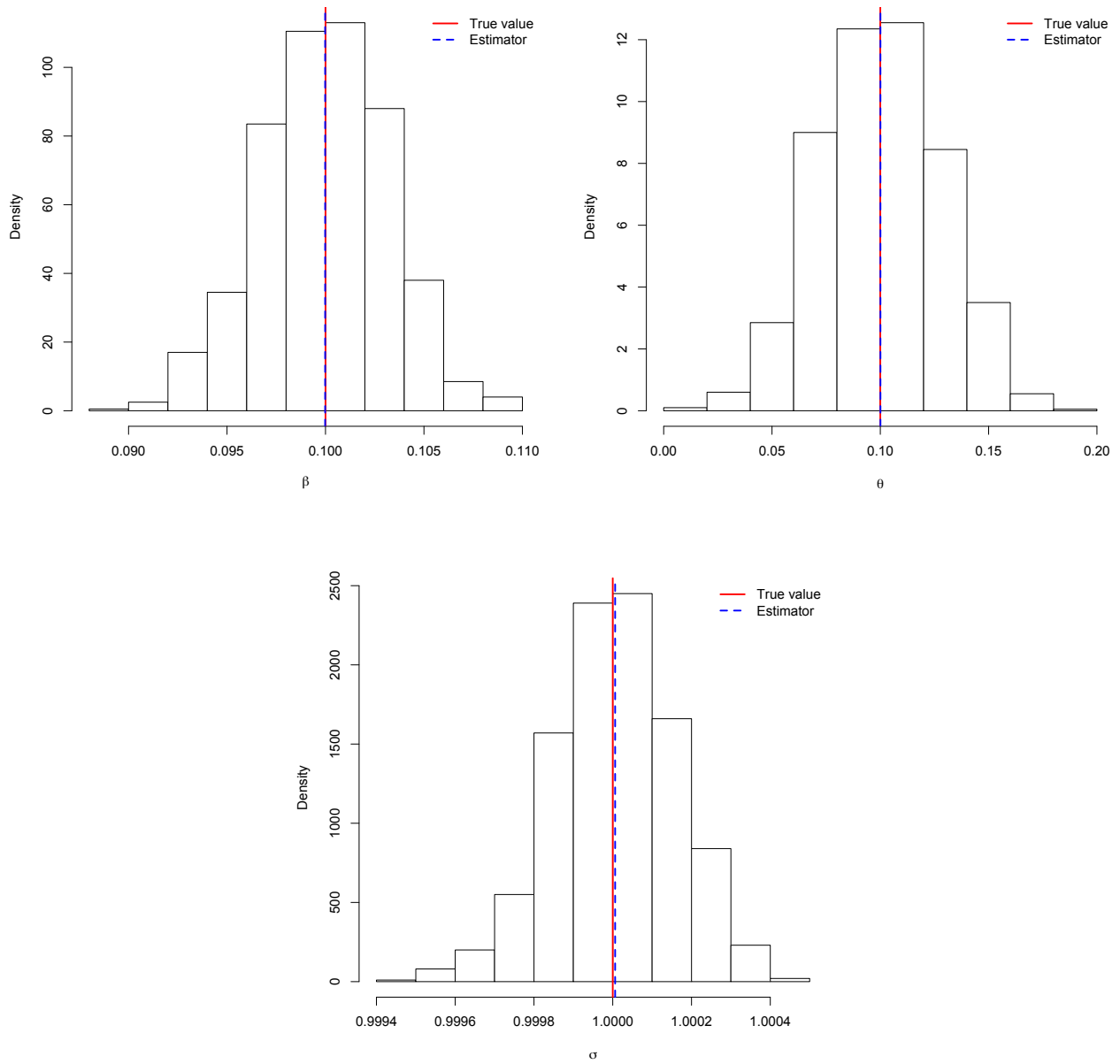


Figure 6 Histograms of 1000 estimators of β , γ and θ for data sets with $M = 50000$ and $\delta = 1.0 \times 10^{-7}$ obtained by the MLE and quadratic variation. The parameter values were $\beta = 0.1$, $\theta = 0.1$ and $\sigma = 1$.



Appendix A. A basic element of the Malliavin-Stein Method

The following result is taken from Theorem 5.2.7 in Nourdin and Peccati (2012)[24]. This is a simplified version of that theorem, specifically highlighting the equivalence between the convergence in law and the Malliavin-based criterion required for our purposes. Here, D denotes the Malliavin derivative.

Theorem 4 (Fourth-moment theorem). *Let $\{F_n\}_{n \geq 1}$ be a sequence of random variables belonging to a fixed Wiener chaos of order $q \geq 2$. Assume that $\lim_{n \rightarrow \infty} \mathbb{E}[F_n^2] = \sigma^2$. Let $Z \sim \mathcal{N}(0, \sigma^2)$ be a Gaussian random variable. Then, the following statements are equivalent:*

- i) $F_n \xrightarrow{\mathcal{L}} Z$ as $n \rightarrow \infty$ (convergence in distribution).
- ii) $\text{Var}(\|DF_n\|_S^2) \rightarrow 0$ as $n \rightarrow \infty$.

Remark 2. *It should be noted that the Malliavin-Stein method is much more extensive than the simplified conditions presented in this theorem. While the Fourth-moment theorem provides a powerful "shortcut" for proving asymptotic normality within a fixed chaos, the full Malliavin-Stein framework allows for the calculation of explicit, non-asymptotic bounds on the distance between distributions (such as Total Variation or Wasserstein distances) for much more general functionals of Gaussian fields.*

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